

Time Series Analysis

Chapter 2: ARMA models



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Acronyms used in this material

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model)
- **ARIMA**: AutoRegressive Integrated Moving Average (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **EU**: European Union
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GDP**: Gross Domestic Product
- **HICP**: Harmonised Index of Consumer Prices
- **INS**: Institutul Național de Statistică — the National Institute of Statistics (Romania)
- **LB**: Ljung–Box (test)
- **MA**: Moving Average
- **MLE**: Maximum Likelihood Estimation
- **MSE**: Mean Squared Error
- **OLS**: Ordinary Least Squares
- **OPG**: Outer Product of Gradients
- **PACF**: Partial AutoCorrelation Function
- **QQ**: Quantile–Quantile (plot)
- **VAR**: Vector AutoRegression

Today's question and route

- **Question**: a stationary series is correlated with its own past; which small model captures this memory and turns it into forecasts with honest uncertainty?
  - the answer of Box and Jenkins: **ARMA(p, q)** models, a few parameters instead of infinitely many Wold weights
- **Route** of the chapter
  - AR(p) and MA(q): stationarity, invertibility, characteristic roots, ACF and PACF
  - ARMA(p, q): the causal MA(∞) form, the impulse responses, identification
  - estimation (Yule–Walker, conditional least squares, maximum likelihood), AIC and BIC, residual diagnostics
  - forecasting and the Box–Jenkins method on Romanian GDP, inflation, BET, EUR/RON and the sunspots
- Seminar 2 comes before this lecture: it gives the formulas; here we derive them and use them on data

Learning outcomes

- Check the stationarity of an AR(p) and the invertibility of an MA(q) from the roots of their polynomials
- Derive the mean, the autocovariances and the ψ weights of AR(1), AR(2), MA(q) and ARMA(1,1)
- Propose candidate orders from the ACF and the PACF, and choose among them with AIC and BIC
- Estimate ARMA models by Yule–Walker, conditional least squares and maximum likelihood, and read the output
- Test the residuals: Ljung–Box with $m - p - q$ degrees of freedom, Jarque–Bera
- Compute ARMA forecasts and their intervals, and explain mean reversion

Reading and tools

- Textbook: Huang and Petukhina (2022), Ch. 3–4 (stationary models, ARMA)
  - companion, free online: Hyndman and Athanasopoulos (2021), Ch. 9 (ARIMA models)
- Theory: Brockwell and Davis (2016), Ch. 3 (ARMA) and Ch. 5 (estimation); Hamilton (1994), Ch. 3–5; the classic: Box, Jenkins and Reinsel (2008)
- Python Quantlets of this chapter: Quantlets/Ch_02
  - ARMA estimation with `statsmodels` (`ARIMA`, `ArmaProcess`, `acorr_ljungbox`)
- Lecture notebook: open in Google Colab
- Video course: Applied Time Series Analysis with Python

Tools from Chapter 1

- **Weak stationarity**: constant mean μ and autocovariance $\gamma(h) = \mathrm{Cov}(X_t, X_{t+h})$ that depends only on the lag h
  - ACF $\rho(h) = \gamma(h)/\gamma(0)$; PACF ϕ_{hh} = correlation of X_t and X_{t+h} after removing $X_{t+1}, \ldots, X_{t+h-1}$
- **White noise** $\varepsilon_t \sim \mathrm{WN}(0, \sigma^2)$: mean 0, variance σ^2, no autocorrelation
  - the **lag operator**: $LX_t = X_{t-1}$, $L^k X_t = X_{t-k}$; polynomials in L act on series
- **Wold decomposition** (Wold (1938)): every stationary, purely nondeterministic process is $X_t = \mu + \sum_{j \geq 0} \psi_j \varepsilon_{t-j}$, $\psi_0 = 1$, $\sum_j \psi_j^2 < \infty$
  - details and charts: Chapter 1, sections on the lag operator, Wold and the sample ACF and PACF

Parsimonious representation of the Wold weights

- Wold says **what** a stationary process looks like, not **how to estimate it**
  - $\psi_1, \psi_2, \ldots$: infinitely many unknowns, and only T observations
- **Idea**: write $\psi(L) = \sum_j \psi_j L^j$ as a **ratio of two short polynomials**
  - $\psi(L) = \theta(L)/\phi(L)$ with $\phi(L) = 1 - \phi_1 L - \cdots - \phi_p L^p$, $\theta(L) = 1 + \theta_1 L + \cdots + \theta_q L^q$
  - then $\phi(L)(X_t - \mu) = \theta(L)\varepsilon_t$: the **ARMA$(p, q)$** model with $p + q + 2$ parameters (ϕ, θ, μ, σ^2)
- A rational function approximates almost any smooth $\psi(L)$ well with small p and q
  - the principle of **parsimony**: the smallest model that leaves white-noise residuals

AR, MA and ARMA models

- **AR(p)**, autoregressive: today depends on the last p values of the series
  - $X_t = c + \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \varepsilon_t$; origin: Yule (1927), sunspots
- **MA(q)**, moving average: today depends on the last q shocks
  - $X_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$; origin: Slutzky (1937), sums of random causes
- **ARMA(p, q)**: both parts; popularised as a complete method by Box, Jenkins and Reinsel (2008) (first edition 1970)
  - c is the **intercept**; the mean is $\mu = c/(1 - \phi_1 - \cdots - \phi_p)$, not c
- Throughout: $\varepsilon_t \sim \text{WN}(0, \sigma^2)$, and for likelihood and intervals ε_t i.i.d. $N(0, \sigma^2)$ (independent and identically distributed)

The AR(1) model

- $X_t = c + \phi X_{t-1} + \varepsilon_t$, or $(1 - \phi L)X_t = c + \varepsilon_t$
  - ϕ: the share of today's deviation from the mean that survives until tomorrow
- **Back substitution** k times:
  - $X_t = c(1 + \phi + \cdots + \phi^{k-1}) + \phi^k X_{t-k} + \sum_{j=0}^{k-1} \phi^j \varepsilon_{t-j}$
  - if $|\phi| < 1$: $\phi^k \to 0$ and $X_t = \dfrac{c}{1-\phi} + \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j}$
- This is the Wold form with $\psi_j = \phi^j$: the **causal** (MA(∞)) representation
  - causal = X_t depends only on present and past shocks
  - $\phi = 1$: random walk (Chapter 1), no stationary solution; $|\phi| > 1$: explosive, or a solution that depends on future shocks, excluded

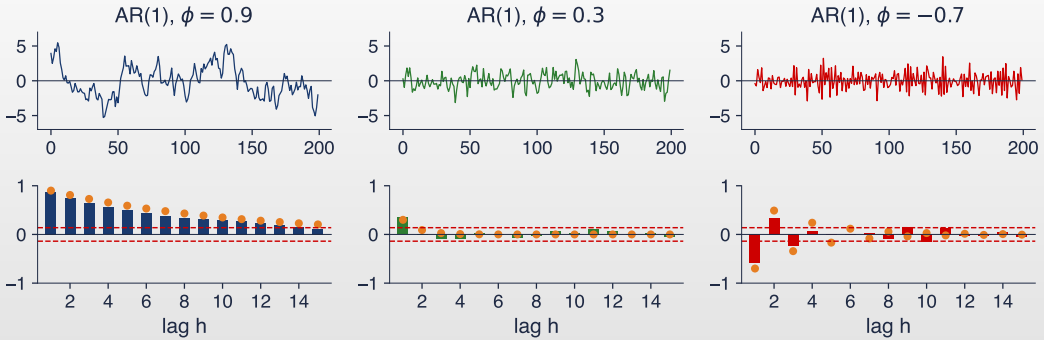
Moments of a stationary AR(1)

- **Mean**: take expectations, $\mu = c + \phi\mu$, so $\mu = c/(1-\phi)$
  - stationarity is used here: $E[X_t] = E[X_{t-1}]$
- **Variance**: $\gamma(0) = \phi^2\gamma(0) + \sigma^2$, since ε_t is uncorrelated with X_{t-1}
  - $\gamma(0) = \sigma^2/(1-\phi^2)$: grows without bound as $|\phi| \to 1$
- **Autocovariances**: multiply $X_t - \mu = \phi(X_{t-1} - \mu) + \varepsilon_t$ by $X_{t-h} - \mu$, $h \geq 1$, and take expectations
  - $\gamma(h) = \phi\gamma(h-1)$, so $\gamma(h) = \phi^h\gamma(0)$ and $\rho(h) = \phi^h$
  - geometric decay; alternating signs if $\phi < 0$
- **Half-life** of a shock: the h with $\phi^h = 1/2$, i.e. $h = \ln 0.5 / \ln \phi$

Worked example: an AR(1)

- $X_t = 2 + 0.8X_{t-1} + \varepsilon_t$, $\sigma^2 = 1$
- Stationary? $|\phi| = 0.8 < 1$: yes
  - mean $\mu = 2/(1 - 0.8) = 10$; the intercept is 2, the mean is 10
- Variance and ACF
  - $\gamma(0) = 1/(1 - 0.64) = 2.78$
  - $\rho(3) = 0.8^3 = 0.512$
- Half-life $= \ln 0.5 / \ln 0.8 = 3.11$ periods
  - after about 3 periods, half of a shock is gone
- ψ weights: $1, 0.8, 0.64, 0.512, \ldots$: a shock of 1 today moves X_{t+2} by 0.64

AR(1) for three values of ϕ



- The same 200 shocks, $\phi = 0.9$, 0.3, -0.7; bottom: sample ACF (bars) and $\rho(h) = \phi^h$ (dots)

TSA_ch2_ar_models

Interpreting the AR(1) paths

- $\phi = 0.9$: long swings away from 0; sample $\hat{\rho}(1) = 0.88$, variance 4.98 (theory 5.26)
  - strong persistence looks like a trend over short windows
- $\phi = 0.3$: close to white noise; only $\hat{\rho}(1) = 0.35$ is clearly outside the band
  - variance 1.18 (theory 1.10)
- $\phi = -0.7$: zig-zag; $\hat{\rho}(1) = -0.58$, $\hat{\rho}(2) = 0.34$, alternating signs
  - in economics: overshooting and correction, or a measurement error that is corrected next period
- With $T = 200$ the sample ACF is below the theoretical one at long lags: a small-sample bias that also affects the estimators (Section 5)

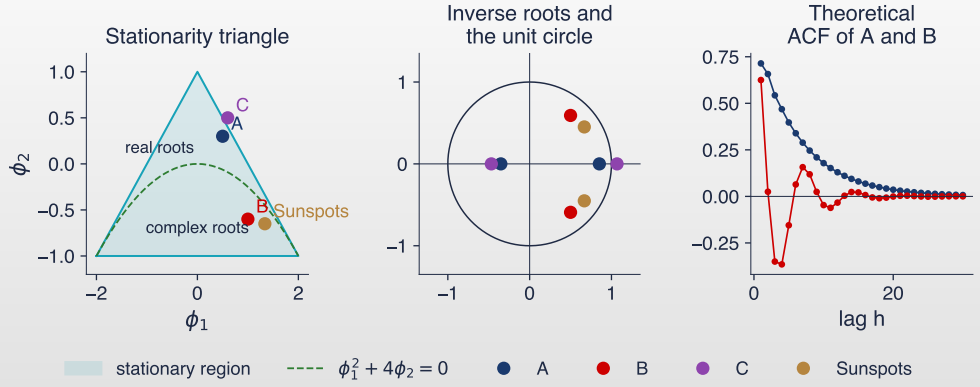
The AR(p) model and its characteristic polynomial

- $\phi(L)X_t = c + \varepsilon_t$, with $\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$, a polynomial in a complex number z
  - factorise: $\phi(z) = (1 - \lambda_1 z)\cdots(1 - \lambda_p z)$, where $z_i = 1/\lambda_i$ are the roots
- **Stationarity (causality) condition**: all roots of $\phi(z) = 0$ lie **outside the unit circle**, $|z_i| > 1$
  - equivalently: all **inverse roots** λ_i lie inside, $|\lambda_i| < 1$ (the form shown by software)
  - reason: $1/(1 - \lambda L) = \sum_j \lambda^j L^j$ converges only if $|\lambda| < 1$; each factor is an AR(1)
- A root on the circle ($|z| = 1$, e.g. $z = 1$) is a **unit root**: non-stationarity, the topic of Chapter 3
  - quick necessary check: $\phi_1 + \cdots + \phi_p < 1$ (i.e. $\phi(1) > 0$)

AR(2): the stationarity triangle

- $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$ is stationary if and only if
  - $\phi_1 + \phi_2 < 1, \quad \phi_2 - \phi_1 < 1, \quad |\phi_2| < 1$
- The roots are complex when $\phi_1^2 + 4\phi_2 < 0$
  - then the ACF is a **damped cosine**: pseudo-cycles of average length $2\pi / \arccos\left(\phi_1 / (2\sqrt{-\phi_2})\right)$
  - damping factor per period: $\sqrt{-\phi_2}$, the modulus of the inverse roots
- **Yule–Walker equations** for the ACF: multiply by X_{t-h}, take expectations, divide by $\gamma(0)$
  - $\rho(1) = \phi_1 + \phi_2 \rho(1)$, so $\rho(1) = \phi_1 / (1 - \phi_2)$
  - $\rho(h) = \phi_1 \rho(h-1) + \phi_2 \rho(h-2)$ for $h \geq 2$: the ACF obeys the same recursion as the process

Three AR(2) processes and the sunspots



- A: $\phi = (0.5, 0.3)$; B: $(1.0, -0.6)$; C: $(0.6, 0.5)$; Sunspots: Yule's AR(2), estimated in Section 10
- Middle: inverse roots λ_i; stationary means inside the circle

TSA_ch2_ar_models

Interpreting the AR(2) examples

- **A**: real roots -2.84 and 1.17, both outside the circle: stationary
  - $\rho(1) = 0.5/0.7 = 0.714$, $\rho(2) = 0.5\rho(1) + 0.3 = 0.657$: slow, smooth decay
- **B**: complex roots $0.833 \pm 0.986\,i$, modulus $1.291 > 1$: stationary
  - ACF: a wave with period 7.2 lags, damped by 0.775 per lag: a **stochastic cycle**
- **C**: $\phi_1 + \phi_2 = 1.1 > 1$; one root $0.936 < 1$: **not** stationary (explosive)
  - each coefficient is below 1, yet the process explodes: always check the roots, not the coefficients one by one
- Complex roots are how linear models produce business cycles and the 11-year solar cycle

The PACF of an AR(p) cuts off after lag p

- ϕ_{hh} = the last coefficient in the best linear prediction of X_t from $X_{t-1}, \ldots, X_{t-h}$ (Chapter 1)
  - computed from the ACF by the Durbin–Levinson recursion (Durbin (1960))
- For an AR(p) and $h > p$, the best predictor is the model itself: $\phi_1 X_{t-1} + \cdots + \phi_p X_{t-p}$
  - so $\phi_{hh} = 0$ for $h > p$, and $\phi_{pp} = \phi_p$
- **Identification rule**: ACF decays gradually, PACF has p spikes and then stays inside the band: AR(p)
  - band for the sample PACF of an AR(p) at lags $h > p$: $\pm 1.96/\sqrt{T}$

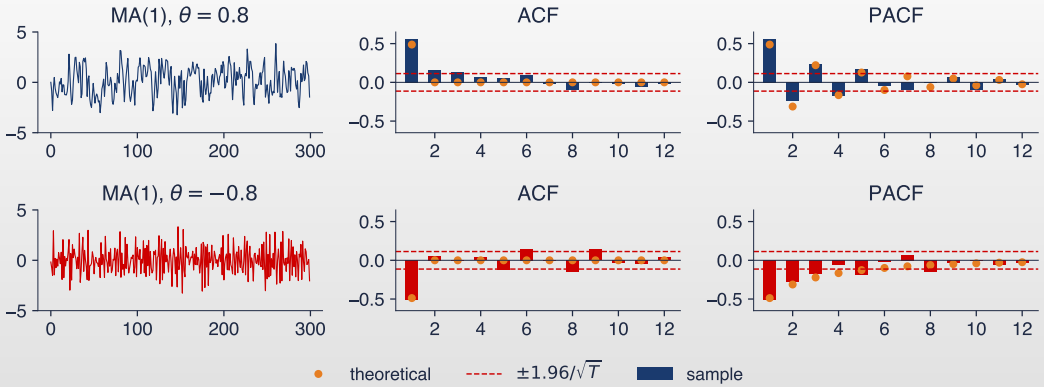
Recap: Autoregressive models

- AR(p): $\phi(L)X_t = c + \varepsilon_t$; stationary iff all roots of $\phi(z)$ lie outside the unit circle
- Mean $\mu = c/\phi(1)$; ACF from the Yule–Walker recursion, a mix of geometric decays and damped waves
- AR(1): $\rho(h) = \phi^h$, $\psi_j = \phi^j$, half-life $\ln 0.5 / \ln \phi$
- PACF cuts off after lag p: the signature of an AR(p)

The MA(q) model

- $X_t = \mu + \theta(L)\varepsilon_t = \mu + \varepsilon_t + \theta_1\varepsilon_{t-1} + \cdots + \theta_q\varepsilon_{t-q}$
  - a finite Wold sum: **always stationary**, for any θ
- Moments (derived in Seminar 1 and Chapter 1): with $\theta_0 = 1$
  - $E[X_t] = \mu$; $\gamma(h) = \sigma^2 \sum_{j=0}^{q-h} \theta_j\theta_{j+h}$ for $0 \le h \le q$; $\gamma(h) = 0$ for $h > q$
- **Identification rule**: the ACF **cuts off** after lag q; the PACF decays gradually
  - the mirror image of the AR(p) rule
  - band for the sample ACF at lags $h > q$ (Bartlett): $\pm 1.96\sqrt{(1 + 2\sum_{j=1}^{q} \hat{\rho}(j)^2)/T}$, wider than $\pm 1.96/\sqrt{T}$

MA(1): $\theta = 0.8$ and $\theta = -0.8$



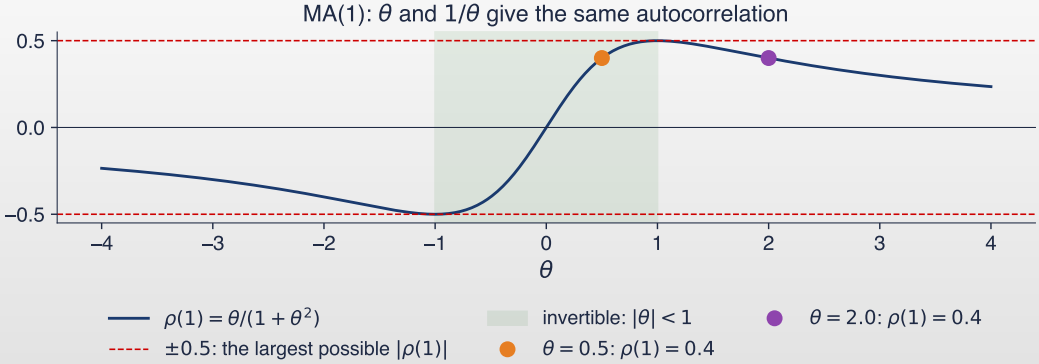
- $T = 300$; the ACF has one spike, the PACF decays (with alternating signs when $\theta > 0$)

TSA_ch2_ma_models

Interpreting the MA(1) correlograms

- Theory: $\rho(1) = \theta/(1+\theta^2) = \pm 0.488$, $\rho(h) = 0$ for $h \geq 2$
  - sample: $\hat{\rho}(1) = 0.56$ and -0.51
- The PACF does not cut off: $\hat{\phi}_{22} = -0.23$ for $\theta = 0.8$
  - an MA(1) is an AR(∞) (next slides), so every past value helps a little
- An MA(1) remembers exactly one period: a shock today affects tomorrow and is then forgotten
- Economic examples: overlapping observations (Section 9), a measurement error, a bid–ask bounce in prices

The largest autocorrelation of an MA(1)



⊡ $\rho(1) = \theta/(1+\theta^2)$ as a function of θ

TSA_ch2_ma_models

Interpreting the curve $\rho(1)$

- Maximum $|\rho(1)| = 0.5$ at $\theta = \pm 1$: an MA(1) can never produce $|\rho(1)| > 0.5$
  - if $\hat{\rho}(1) = 0.7$ and then nothing: not an MA(1); try an AR or an MA(q) with more terms
- $\theta = 0.5$ and $\theta = 2$ give the same $\rho(1) = 0.4$
  - in general θ and $1/\theta$ give the same ACF (with σ^2 rescaled by θ^2)
  - the data cannot tell them apart: an **identification problem**
- The convention that removes the ambiguity: invertibility, $|\theta| < 1$ (next slide)

Invertibility

- MA(1): $\varepsilon_t = X_t - \theta\varepsilon_{t-1}$; substitute repeatedly ($\mu = 0$):
  - $\varepsilon_t = X_t - \theta X_{t-1} + \theta^2 X_{t-2} - \cdots = \sum_{j\geq 0}(-\theta)^j X_{t-j}$, if $|\theta| < 1$
  - an **AR(∞) representation**: $X_t = \theta X_{t-1} - \theta^2 X_{t-2} + \cdots + \varepsilon_t$
- **Invertible**: the shocks can be recovered from the past of X; condition: all roots of $\theta(z) = 0$ outside the unit circle
  - MA(1): $|\theta| < 1$; the same algebra as AR stationarity, applied to $\theta(z)$
- Why we want it
  - uniqueness: among θ and $1/\theta$ only one is invertible
  - forecasting: the shocks ε_t are computed from the data, so forecasts can use them
  - software reports invertible estimates; a root near the circle signals over-differencing (Chapter 1)

Worked example: $\theta = 2$ or $\theta = 0.5$?

- $X_t = \varepsilon_t + 2\varepsilon_{t-1}$, $\sigma^2 = 1$; and $Y_t = u_t + 0.5u_{t-1}$, $\mathrm{Var}(u_t) = 4$
- X: $\gamma(0) = 1 \cdot (1 + 4) = 5$, $\gamma(1) = 2 \cdot 1 = 2$
  - Y: $\gamma(0) = 4 \cdot (1 + 0.25) = 5$, $\gamma(1) = 0.5 \cdot 4 = 2$
- Same mean, same autocovariances: with Gaussian shocks, the **same distribution**
  - no amount of data separates them
- Only Y is invertible: $u_t = Y_t - 0.5Y_{t-1} + 0.25Y_{t-2} - \ldots$
  - for X, the same sum with weights $(-2)^j$ explodes
- Rule: report the invertible version ($\theta = 0.5$, $\sigma^2 = 4$)

Recap: Moving-average models

- MA(q) is always stationary; its ACF cuts off after lag q, its PACF decays
- MA(1): $\rho(1) = \theta/(1+\theta^2)$, at most 0.5 in absolute value
- Invertible iff the roots of $\theta(z)$ lie outside the unit circle; then MA = AR(∞)
- θ and $1/\theta$ are observationally equivalent; we keep the invertible one

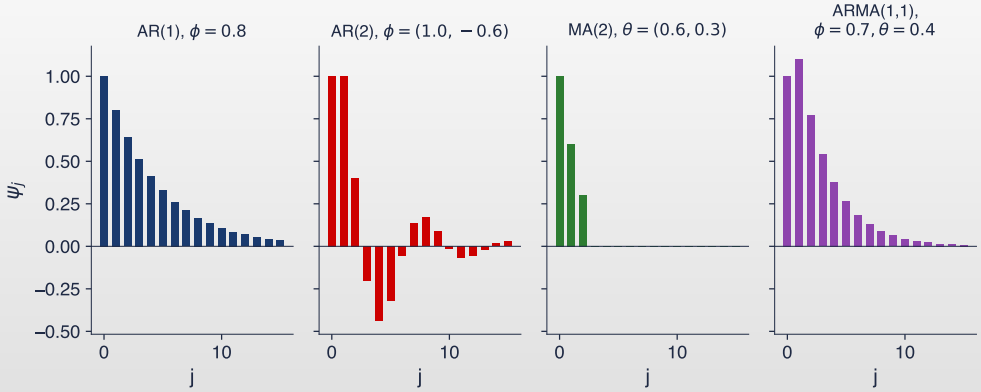
The ARMA(p, q) model

- $\phi(L)(X_t - \mu) = \theta(L)\varepsilon_t$, i.e.
  - $X_t = c + \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$, $c = \mu\,\phi(1)$
- Three conditions (Brockwell and Davis (2016), Ch. 3)
  - **stationary and causal**: roots of $\phi(z)$ outside the unit circle; then $X_t - \mu = \psi(L)\varepsilon_t$, $\psi(L) = \theta(L)/\phi(L)$
  - **invertible**: roots of $\theta(z)$ outside the unit circle; then $\pi(L)(X_t - \mu) = \varepsilon_t$, $\pi(L) = \phi(L)/\theta(L)$
  - **no common factors**: $\phi(z)$ and $\theta(z)$ share no root
- ARMA(1,1) can match an ACF that decays at rate ϕ from a first value set by θ (below ϕ if $\theta < 0$, above if $\theta > 0$); a pure AR or MA would need many terms for it

The ψ weights and the impulse response

- From $\phi(L)\psi(L) = \theta(L)$, match the coefficients of L^j:
  - $\psi_0 = 1, \quad \psi_j = \theta_j + \sum_{i=1}^{\min(j,p)} \phi_i \psi_{j-i} \quad (\theta_j = 0 \text{ for } j > q)$
- ARMA(1,1): $\psi_1 = \phi + \theta$, $\psi_j = \phi^{j-1}(\phi + \theta)$ for $j \geq 1$
  - example $\phi = 0.7$, $\theta = 0.4$: $\psi_1 = 1.10$, $\psi_2 = 0.77$
- **Impulse response**: $\psi_j = \partial X_{t+j} / \partial \varepsilon_t$, the effect of a unit shock after j periods
  - the same object drives the forecast errors (Section 7) and the impulse responses of VAR models (Chapter 6)

Impulse responses of four models



- ψ_j, $j = 0, \ldots, 15$, from the recursion above

TSA_ch2_arma_acf

Interpreting the impulse responses

- AR(1): geometric decay 0.8^j; the shock never vanishes completely, but fades fast enough for $\sum \psi_j^2 < \infty$
- AR(2) with complex roots: the response changes sign, $\psi_3 = -0.20$, $\psi_4 = -0.44$: a shock creates a cycle
- MA(2): exactly zero after two periods: finite memory
- ARMA(1,1): the MA term lifts the first response to $\psi_1 = 1.10 > 1$; afterwards the AR part takes over

Moments of the ARMA(1,1)

- $X_t = \phi X_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}$, $|\phi| < 1$, $|\theta| < 1$, $\phi + \theta \neq 0$
- Variance, from $\gamma(0) = \sigma^2 \sum_j \psi_j^2$:
  - $\gamma(0) = \sigma^2 \dfrac{1 + 2\phi\theta + \theta^2}{1 - \phi^2}$; for $\phi = 0.7$, $\theta = 0.4$, $\sigma^2 = 1$: 3.37
- Autocorrelations:
  - $\rho(1) = \dfrac{(1 + \phi\theta)(\phi + \theta)}{1 + 2\phi\theta + \theta^2}$, $\quad \rho(h) = \phi\, \rho(h-1)$ for $h \geq 2$
  - example: $\rho(1) = 0.819$, $\rho(2) = 0.573$
- The ACF decays like an AR(1) from lag 1 onwards, but $\rho(1) \neq \phi$: the MA part only moves the starting point

Common factors: a model that is too large

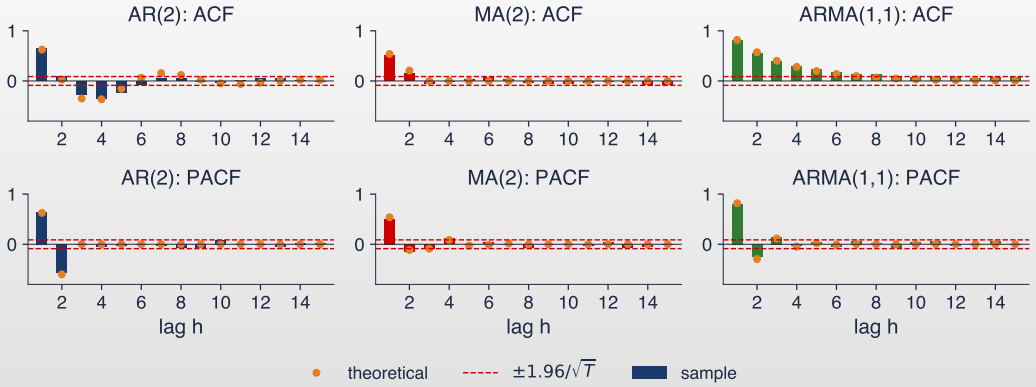
- $X_t = 0.5X_{t-1} + \varepsilon_t - 0.5\varepsilon_{t-1}$
  - $(1 - 0.5L)X_t = (1 - 0.5L)\varepsilon_t$: cancel the factor, $X_t = \varepsilon_t$, white noise
- In general $\phi = -\theta$ in an ARMA(1,1) gives white noise, for any value
  - the likelihood is flat along the line $\phi = -\theta$: the parameters are not identified
- Symptoms in software output
  - large and opposite estimates, e.g. $\hat{\phi} = 0.9$, $\hat{\theta} = -0.88$, with huge standard errors
  - a log-likelihood no better than the smaller model
- Remedy: reduce p and q together; prefer the smallest model with white-noise residuals

Identification: the ACF–PACF table

Model	ACF	PACF
White noise	no significant lag	no significant lag
AR(p)	decays (geometric or damped wave)	**cuts off** after lag p
MA(q)	**cuts off** after lag q	decays
ARMA(p, q)	decays after lag $q - p$	decays after lag $p - q$
Unit root (Ch. 3)	very slow, almost linear decay	$\hat{\phi}_{11} \approx 1$

- The table proposes **candidates**; the information criteria and the diagnostics decide (Sections 5–6)
- Mixed ARMA models are hard to spot by eye: both functions just decay

Theoretical and sample ACF and PACF



- AR(2) $\phi = (1.0, -0.6)$, MA(2) $\theta = (0.6, 0.3)$, ARMA(1,1) $\phi = 0.7$, $\theta = 0.4$; one simulated path of $T = 500$ each

TSA_ch2_arma_acf

Interpreting the three patterns

- AR(2): the PACF has two spikes, $\phi_{22} = \phi_2 = -0.60$, then zero; the ACF is a damped wave
  - identification is easy: read p from the PACF
- MA(2): $\rho(1) = 0.538$, $\rho(2) = 0.207$, then zero; the PACF decays slowly
  - the sample ACF at lag 2 is only just outside the band: with real data, MA(1) and MA(2) can look alike
- ARMA(1,1): both decay; the ACF looks like an AR(1), the PACF has a second, smaller spike
  - an analyst who reads only the PACF would choose AR(2) or AR(3): close, but more parameters
- Sample correlograms are noisy even with $T = 500$; with 100 quarterly observations, much more so

Recap: ARMA models

- ARMA(p, q): $\phi(L)(X_t - \mu) = \theta(L)\varepsilon_t$; causal if $\phi(z)$ has roots outside the circle, invertible if $\theta(z)$ has
- ψ weights by the recursion $\psi_j = \theta_j + \sum \phi_i \psi_{j-i}$: the impulse response
- No common factors; a flat likelihood and opposite estimates signal one
- ACF cuts off: MA; PACF cuts off: AR; both decay: ARMA

Three estimators

- **Yule–Walker** (method of moments, Yule (1927); Walker (1931)): replace $\rho(h)$ by $\hat{\rho}(h)$ in the Yule–Walker equations
  - pure AR only; fast; always gives a stationary model; less precise when roots are near the circle
- **Conditional least squares**: minimise $\sum_t \varepsilon_t^2$, treating the first values as fixed
  - AR(p): ordinary least squares (OLS) of X_t on $1, X_{t-1}, \ldots, X_{t-p}$
  - MA and ARMA: the residuals $\varepsilon_t = X_t - \mu - \theta\varepsilon_{t-1} - \ldots$ are computed recursively from $\varepsilon_0 = 0$; numerical minimisation
- **Exact Gaussian maximum likelihood** (MLE): maximise the joint density of $(X_1, \ldots, X_T)$
  - uses the first observations too; the default of `statsmodels` `ARIMA` (Kalman filter, Chapter 10)
- In large samples all three agree for pure AR models; MLE is the reference for MA and ARMA

Yule–Walker for an AR(p)

- For $h = 1, \dots, p$: $\rho(h) = \phi_1 \rho(h-1) + \cdots + \phi_p \rho(h-p)$
  - in matrix form $R\,\phi = \rho$, $R_{ij} = \rho(|i-j|)$; $\phi = (\phi_1, \dots, \phi_p)^\top$, $\rho = (\rho(1), \dots, \rho(p))^\top$
  - R: the $p \times p$ matrix of autocorrelations; the hat marks the sample version
  - estimator $\hat{\phi} = \hat{R}^{-1}\hat{\rho}$; noise variance $\hat{\sigma}^2 = \hat{\gamma}(0)(1 - \hat{\phi}^\top \hat{\rho})$
- AR(2) by Cramer's rule:
  - $\hat{\phi}_1 = \dfrac{\hat{\rho}_1(1-\hat{\rho}_2)}{1-\hat{\rho}_1^2}, \quad \hat{\phi}_2 = \dfrac{\hat{\rho}_2 - \hat{\rho}_1^2}{1-\hat{\rho}_1^2}$
  - $\hat{\phi}_2$ is the sample PACF at lag 2 (Chapter 1)



Gilbert Walker (1868–1958), who extended Yule's method in 1931

Photo: unknown author (1925); public domain; Wikimedia Commons

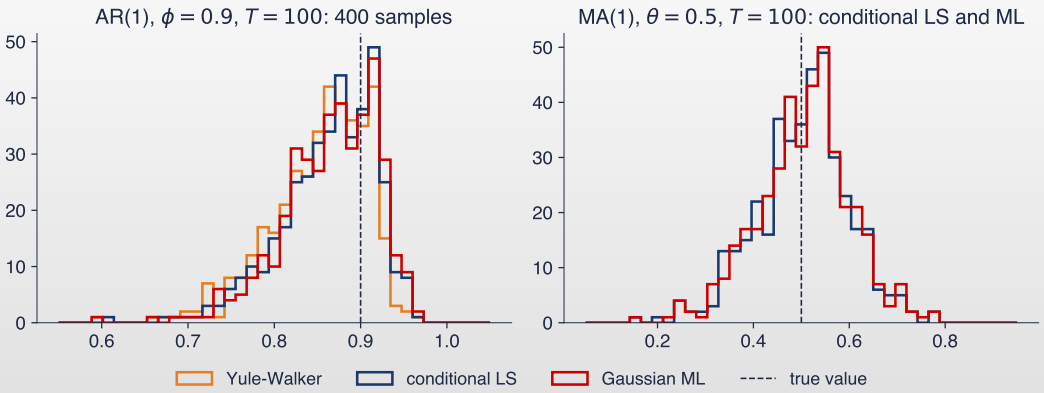
Worked example: Yule–Walker on Romanian GDP growth

- Annual growth of real GDP, 2000Q1–2026Q2 (Section 9): $\hat{\rho}(1) = 0.62$, $\hat{\rho}(2) = 0.40$, $\hat{\gamma}(0) = 18.77$
- Determinant $1 - \hat{\rho}_1^2 = 0.620$
  - $\hat{\phi}_1 = 0.62\,(1 - 0.40)/0.620 = 0.600$
  - $\hat{\phi}_2 = (0.40 - 0.62^2)/0.620 = 0.027$
- $\hat{\sigma}^2 = 18.77\,(1 - \hat{\phi}_1\hat{\rho}_1 - \hat{\phi}_2\hat{\rho}_2) = 11.62$
- $\hat{\phi}_2 \approx 0$: the second lag adds nothing, AR(1) would do
  - but is an AR model the right family at all? Section 9 shows that an MA(3) fits better

Maximum likelihood

- Gaussian likelihood by the **prediction-error decomposition**
  - $f(x_1, \dots, x_T) = \prod_t f(x_t \mid x_{t-1}, \dots, x_1)$; each factor is $N(\hat{x}_{t|t-1}, v_t)$
  - $\ln L = -\frac{1}{2} \sum_{t=1}^{T} \left[\ln(2\pi v_t) + (x_t - \hat{x}_{t|t-1})^2 / v_t \right]$
  - $\hat{x}_{t|t-1}$: one-step prediction; v_t: its error variance, larger for the first observations
- AR(1) by hand: $X_1 \sim N(\mu, \sigma^2/(1-\phi^2))$, then $X_t \mid X_{t-1} \sim N(c + \phi X_{t-1}, \sigma^2)$
  - dropping the first factor gives conditional least squares
- Standard errors (se) from the curvature of $\ln L$; asymptotically
  - AR(1): $\mathrm{se}(\hat{\phi}) \approx \sqrt{(1-\phi^2)/T}$; MA(1): $\mathrm{se}(\hat{\theta}) \approx \sqrt{(1-\theta^2)/T}$
  - t-ratios $\hat{\phi}/\mathrm{se}$ compared with ± 1.96, as in regression
  - `statsmodels` reports by default the **OPG** se (outer product of the scores): the same limit if the model is right, but smaller than the curvature se when the errors have heavy tails; the se in this chapter are OPG

Sampling distributions of the estimators



⊡ 400 simulated samples of $T = 100$ each; left: AR(1) with $\phi = 0.9$; right: MA(1) with $\theta = 0.5$

TSA_ch2_estimation

Interpreting the estimator comparison

- AR(1), $\phi = 0.9$: means 0.853 (Yule–Walker), 0.864 (conditional least squares), 0.866 (MLE)
  - all three are biased downwards; the textbook approximation of the least squares bias is $-(1 + 3\phi)/T = -0.037$
  - Yule–Walker is the most biased near the unit circle
- Standard deviations 0.056, 0.055, 0.054, above the asymptotic 0.044
  - the distribution is skewed to the left: asymptotic intervals are optimistic near $\phi = 1$
- MA(1), $\theta = 0.5$: conditional least squares 0.504 (sd 0.097), MLE 0.506 (sd 0.098); asymptotic sd 0.087
- Practical rule: use MLE; Yule–Walker only for quick starting values or for teaching

Recap: Estimation

- Yule–Walker: moments, pure AR, $\hat{\phi} = \hat{R}^{-1}\hat{\rho}$
- Conditional least squares: OLS for AR, a recursion for MA
- Exact Gaussian MLE: the prediction-error decomposition; the standard in software
- Persistent AR coefficients are biased downwards in small samples

Information criteria

- Fit improves with every parameter; the criteria charge a **penalty** per parameter
  - k = number of estimated parameters (ϕ, θ, μ, σ^2), $\ln L$ = maximised log-likelihood
- **AIC** $= -2 \ln L + 2k$ (Akaike (1974))
  - **AICc** $= \text{AIC} + 2k(k+1)/(T-k-1)$, for small T (Hurvich and Tsai (1989))
- **BIC** $= -2 \ln L + k \ln T$ (Schwarz (1978))
  - $\ln T > 2$ once $T \geq 8$: BIC prefers smaller models
- Choose the model with the **smallest** value; compare only models fitted to the same observations



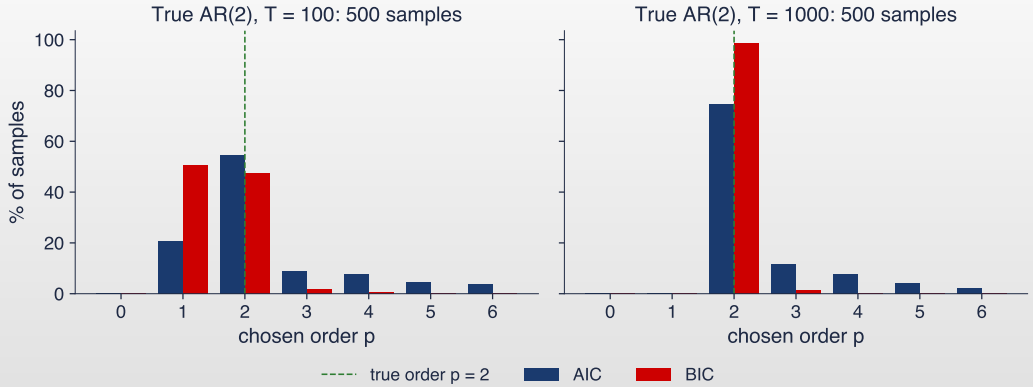
Hirotugu Akaike (1927–2009), Institute of Statistical Mathematics, Tokyo

Photo: The Institute of Statistical Mathematics (2017); CC BY-SA 4.0; Wikimedia Commons

AIC and BIC compared

- **BIC is consistent**: if the true model is among the candidates, BIC finds it with probability $\to 1$
  - but it may choose a model that is too small when T is small
- **AIC targets forecasting**: it estimates the expected out-of-sample fit (Kullback–Leibler distance)
  - it overfits with positive probability, even with large T
- Practice (Hyndman and Athanasopoulos (2021), Ch. 9; Hyndman and Khandakar (2008))
  - automatic searches (`auto.arima`, `pmdarima`) minimise AICc over a grid of (p, q)
  - when AIC and BIC disagree, keep both candidates and let the diagnostics and the forecasts decide

How often AIC and BIC find the true order



- True model AR(2), $\phi = (0.5, 0.25)$; AR(p), $p = 0, \ldots, 6$, fitted by OLS on the same sample; 500 samples per panel

TSA_ch2_selection

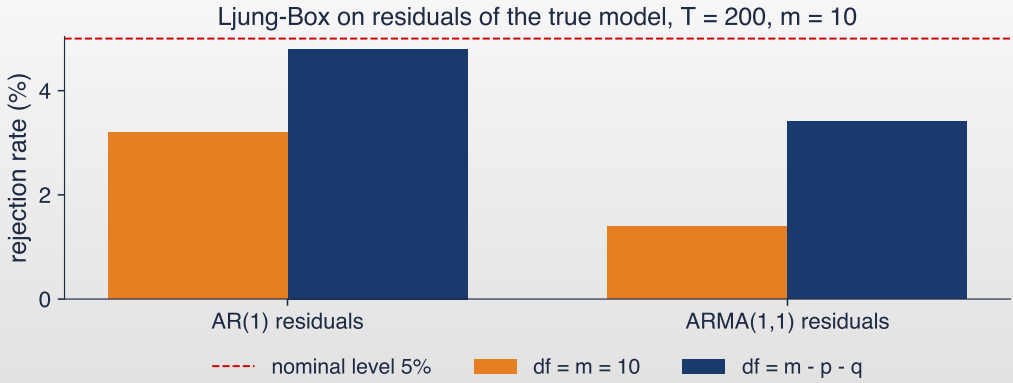
Interpreting the selection experiment

- $T = 100$: AIC finds $p = 2$ in 54.4% of samples, BIC in 47.4%
  - BIC chooses too small a model in 50.4% (the small $\phi_2 = 0.25$ is hard to detect); AIC too large in 24.8%
- $T = 1000$: BIC 98.6%, AIC 74.4%
  - BIC is consistent; AIC still overfits in 25.6% of samples
- Neither criterion is a proof: with quarterly macro data ($T \approx 100$) the order is uncertain
- Overfitting costs little in forecasts; underfitting leaves correlation in the residuals, which the diagnostics catch

Residual diagnostics

- If the model is right, the residuals $\hat{\varepsilon}_t$ behave like white noise
  - plot them: no pattern, no change of variance, no isolated huge values
  - residual ACF inside $\pm 1.96/\sqrt{T}$, except about 1 lag in 20
- **Ljung–Box** on residuals (Ljung and Box (1978)): $Q^*(m) = T(T+2)\sum_{h=1}^{m} \hat{\rho}_{\hat{\varepsilon}}(h)^2/(T-h)$
  - under H_0 (correct model) $Q^*(m) \approx \chi^2(m-p-q)$ (Box and Pierce (1970)): estimating $p+q$ coefficients uses up $p+q$ degrees of freedom
  - choose m well above $p+q$: $m = 10$ for annual or quarterly data, $m = 2s$ for seasonal data (Hyndman and Athanasopoulos (2021))
- **Normality**: Jarque–Bera (Jarque and Bera (1987)) $\text{JB} = \frac{T}{6}\left(S^2 + (K-3)^2/4\right) \approx \chi^2(2)$; QQ plot
  - S = skewness, K = kurtosis (0 and 3 for the Normal distribution); reject normality at 5% if JB > 5.99, the 95% quantile of $\chi^2(2)$
  - non-Normal residuals do not bias $\hat{\phi}$, but they make Normal intervals wrong
- **Squared residuals**: Ljung–Box on $\hat{\varepsilon}_t^2$ detects volatility clustering (Chapter 5)

Justification of the $m - p - q$ degrees of freedom



⊡ Ljung–Box at 5% on the residuals of the **true** model; 1 000 samples of AR(1), half as many of ARMA(1,1)

TSA_ch2_selection

Interpreting the degrees-of-freedom experiment

- With $m = 10$ degrees of freedom the test rejects too rarely: 3.2% (AR(1)) and 1.4% (ARMA(1,1)) instead of 5%
  - residuals are fitted to look uncorrelated at the first lags: their ACF is smaller than that of true white noise
- With $m - p - q$: 4.8% and 3.4%, close to the nominal 5%
  - the correction matters most when $p + q$ is large relative to m
- Software (`acorr_ljungbox`) uses m by default: pass `model_df = p + q`
- Wrong degrees of freedom make a bad model look acceptable

Recap: Model selection and diagnostics

- AIC $= -2 \ln L + 2k$, BIC $= -2 \ln L + k \ln T$; smallest wins; BIC picks smaller models
- BIC is consistent, AIC aims at forecasting and overfits sometimes
- Residuals: plot, ACF, Ljung–Box with $m - p - q$ degrees of freedom, Jarque–Bera, Ljung–Box on squares
- A model passes when its residuals are white noise; normality matters for the intervals

The optimal forecast

- Forecast of X_{T+h} made at time T: $\hat{X}_{T+h|T} = E[X_{T+h} \mid X_T, X_{T-1}, \dots]$
  - it minimises the mean squared error (MSE) among all functions of the past
- Recipe: write the model at $T + h$ and replace
  - future shocks ε_{T+j}, $j \geq 1$, by 0
  - future values X_{T+j} by their forecasts; past values and past residuals stay as they are
- AR(1): $\hat{X}_{T+h|T} = \mu + \phi^h(X_T - \mu)$: **mean reversion** at the speed ϕ
  - MA(q): $\hat{X}_{T+h|T} = \mu$ for $h > q$: after q steps the model knows nothing beyond the mean

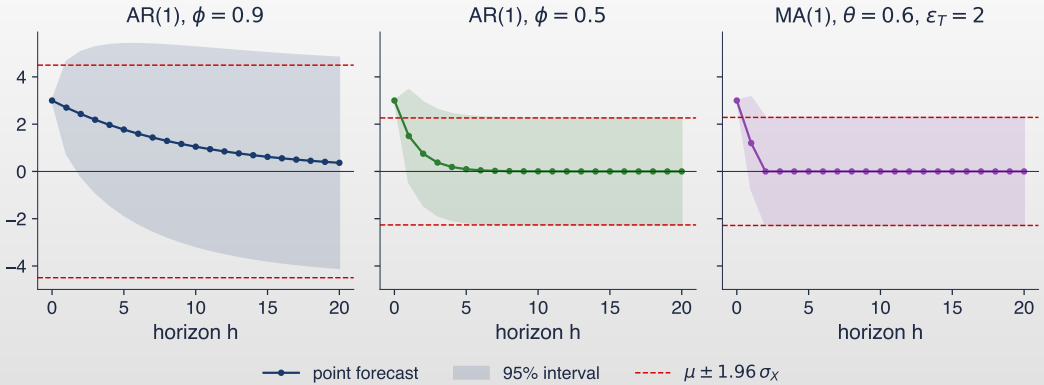
Forecast errors and intervals

- From the causal form $X_{T+h} = \mu + \sum_j \psi_j \varepsilon_{T+h-j}$:
  - error $e_{T+h} = X_{T+h} - \hat{X}_{T+h|T} = \varepsilon_{T+h} + \psi_1 \varepsilon_{T+h-1} + \cdots + \psi_{h-1} \varepsilon_{T+1}$
  - variance $\sigma_h^2 = \sigma^2(1 + \psi_1^2 + \cdots + \psi_{h-1}^2)$, increasing in h
- **95% interval**: $\hat{X}_{T+h|T} \pm 1.96\,\sigma_h$ (Gaussian shocks)
  - $h = 1$: $\pm 1.96\,\sigma$; $h \to \infty$: $\sigma_h^2 \to \gamma(0)$, the unconditional variance
- What the interval leaves out
  - parameter uncertainty ($\hat{\phi}$ instead of ϕ), model uncertainty, fat tails: real coverage is usually below 95%
- Errors of forecasts made at the same time for horizons h and $h+1$ are correlated: they share shocks

Worked example: forecasting an AR(1)

- $X_t = 2 + 0.8X_{t-1} + \varepsilon_t$, $\sigma^2 = 1$, $\mu = 10$; last value $X_T = 12$
- Point forecasts
  - $\hat{X}_{T+1} = 10 + 0.8 \cdot 2 = 11.60$; $\hat{X}_{T+2} = 10 + 0.64 \cdot 2 = 11.28$; $\hat{X}_{T+10} = 10.21$
- Error variances: $\sigma_1^2 = 1$, $\sigma_2^2 = 1 + 0.8^2 = 1.64$
  - 95% intervals: $h = 1$: $[9.64, 13.56]$; $h = 2$: $[8.77, 13.79]$
- Long run: forecast $\to 10$, standard deviation $\to \sqrt{\gamma(0)} = 1.67$

Mean reversion in forecasts



- Start at $X_T = 3$, $\mu = 0$, $\sigma = 1$; dashed: the unconditional band $\mu \pm 1.96\,\sigma_X$

TSA_ch2_forecasting

Interpreting the forecast paths

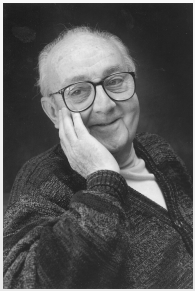
- $\phi = 0.9$: slow reversion; after 5 periods the forecast is still 1.77, standard error 1.85
  - the interval approaches $\pm 1.96 \times 2.29$ only after many periods
- $\phi = 0.5$: after 5 periods the forecast is 0.094, practically the mean
- MA(1): one informative step ($\theta \varepsilon_T = 1.2$), then the mean; the interval jumps to its long-run width at $h = 2$
- A stationary model always forecasts a return to the mean: useful for spreads and growth rates, misleading if the mean shifts

Recap: Forecasting

- $\hat{X}_{T+h|T}$: replace future shocks by 0 and future values by forecasts
- Error variance $\sigma^2 \sum_{j<h} \psi_j^2$; intervals widen up to $\pm 1.96\sqrt{\gamma(0)}$
- Forecasts revert to μ; MA(q) forecasts equal μ beyond q steps
- Intervals ignore parameter and model uncertainty

George Box and Gwilym Jenkins

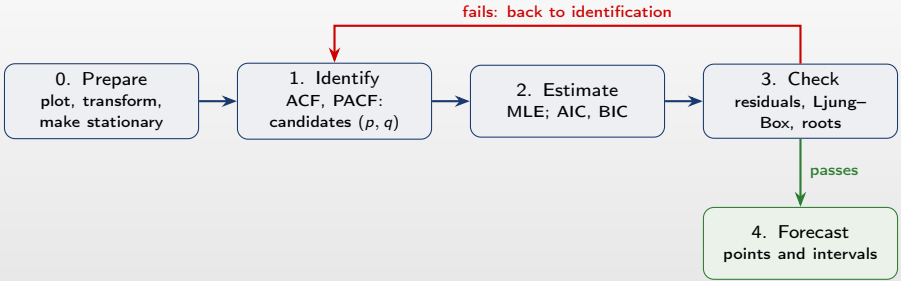
- **George E. P. Box** (1919–2013), statistician, University of Wisconsin–Madison
  - "All models are wrong, but some are useful"
- **Gwilym M. Jenkins** (1932–1982), statistician, Lancaster University
  - *Time Series Analysis: Forecasting and Control*, 1970; fourth edition with Reinsel: Box, Jenkins and Reinsel (2008)
- Their contribution: not new models, but a **method**
  - a loop of identification, estimation and checking that anyone can apply
  - Box–Pierce (1970) and Ljung–Box (1978) come from this programme



George E. P. Box

Photo: DavidMCEddy; CC BY-SA 3.0; Wikimedia Commons

The Box–Jenkins loop



- Step 0 is Chapters 1 and 3: logs, differences, unit-root tests; ARMA needs a stationary series
- The loop ends with the smallest model whose residuals are white noise and whose roots are well inside the admissible region
- Out-of-sample checks (Chapter 0: training and test samples) complete the method

The data

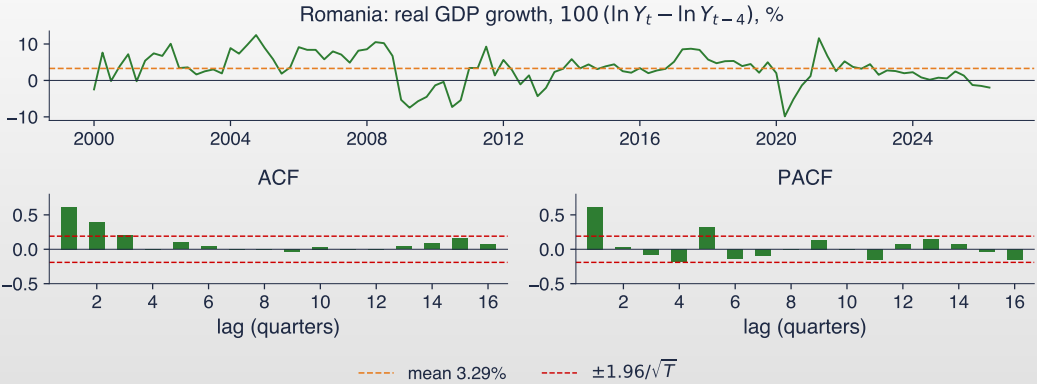
- Romanian real GDP, chain-linked volumes (2010 prices), quarterly, not seasonally adjusted; Eurostat, `namq_10_gdp`
  - the national source: the National Institute of Statistics (INS)
- Annual growth $y_t = 100\,(\ln Y_t - \ln Y_{t-4})$, in %: removes the season (Chapter 1)
  - 2000Q1–2026Q2: $T = 106$ quarters, mean 3.29%, standard deviation 4.35%
  - extremes: -9.9% in 2020Q2, $+12.5\%$ in 2004Q4; last value -2.0%
- Stationary? No trend in the growth rate; a formal test is in Chapter 3



The National Institute of Statistics, Bucharest

Photo: Dan Mihai Pitea (2009); CC BY-SA 3.0; Wikimedia Commons

Step 1: identification



⊡ Annual growth of Romanian real GDP, its ACF and PACF with $\pm 1.96/\sqrt{T} = \pm 0.19$

TSA_ch2_gdp_case

Interpreting the correlograms of GDP growth

- ACF: 0.62, 0.40, 0.21, then -0.01 at lag 4: **cuts off after lag 3**
  - candidate: MA(3)
- PACF: 0.62 at lag 1, then small, except 0.32 at lag 5
  - candidate: AR(1); lag 5 may be chance or a remnant of the season
- Why MA(3)? y_t is the sum of four quarterly growth rates, $y_t \approx g_t + g_{t-1} + g_{t-2} + g_{t-3}$
  - if g_t is close to white noise (Seminar 2, B2), consecutive annual rates share three quarters: an MA(3) by construction
- Ljung–Box $Q^*(8) = 64.9$ on the series itself: strongly autocorrelated, so a model is needed

Step 2: estimation and information criteria

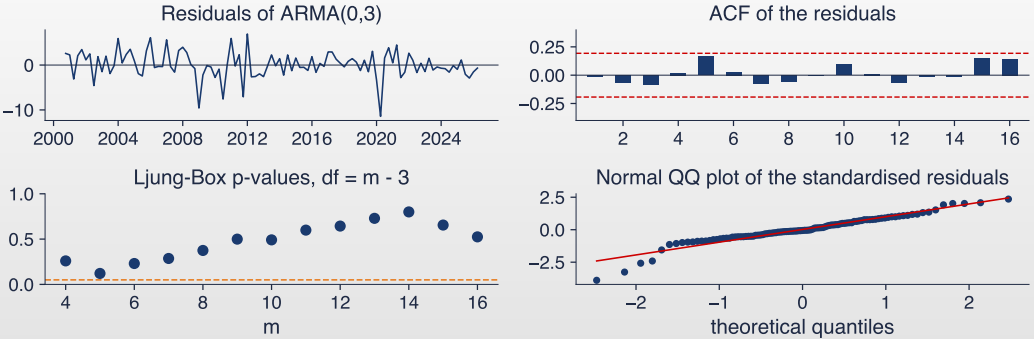
Model	k	$\ln L$	AIC	BIC	$\hat{\sigma}$	LB p-value
ARMA(0,0)	2	−305.83	615.66	620.99	4.33	< 0.001
ARMA(1,0)	3	−279.67	565.33	573.32	3.38	0.023
ARMA(2,0)	4	−279.66	567.32	577.98	3.38	0.037
ARMA(1,1)	4	−279.66	567.32	577.98	3.38	0.013
ARMA(0,3)	5	−270.02	550.04	563.36	3.07	0.375
ARMA(1,3)	6	−268.48	548.97	564.95	3.02	0.648
ARMA(2,3)	7	−267.94	549.89	568.53	3.00	0.543

- Exact Gaussian MLE on the same 106 quarters; LB = Ljung–Box $Q^*(8)$ on the residuals, $8 - p - q$ degrees of freedom; full grid $p, q \leq 3$ in the Quantlet
- **BIC** chooses MA(3); **AIC** chooses ARMA(1,3), lower by only 1.07
- AR(1), the PACF candidate, leaves autocorrelation: LB p-value 0.023

Interpreting the estimated MA(3)

- $\hat{y}_t = 3.17 + \hat{\varepsilon}_t + 0.834\,\hat{\varepsilon}_{t-1} + 0.675\,\hat{\varepsilon}_{t-2} + 0.591\,\hat{\varepsilon}_{t-3}$, $\hat{\sigma} = 3.07$
  - standard errors (OPG): 1.03 (mean), 0.083, 0.102, 0.082: all three MA terms are significant
- The $\hat{\theta}_j$ decline from 0.834 to 0.591: close to, but below, the overlap pattern $(1, 1, 1)$
  - $(1, 1, 1)$ would put roots on the unit circle; the smallest root modulus here is $1.17 > 1$: invertible
- ARMA(1,3): $\hat{\theta}_1 = 1.019 > 1$, yet the roots have modulus at least 1.10: still invertible
  - for $q > 1$, invertibility is read from the roots, not from single coefficients
  - $\hat{\phi} = -0.284$ (se OPG 0.143): one extra parameter for little gain; BIC drops it
- Mean growth 3.17% per year, but with a standard error of 1.03: overlapping data carry less information than T suggests

Step 3: diagnostics of the MA(3)



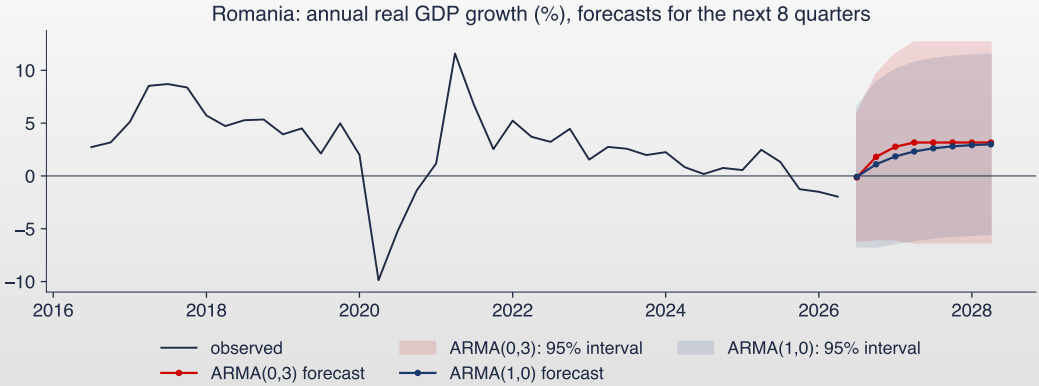
- Residuals, their ACF, Ljung–Box p-values for $m = 4, \ldots, 16$ with $m - 3$ degrees of freedom, Normal QQ plot

TSA_ch2_gdp_case

Interpreting the MA(3) diagnostics

- No residual autocorrelation: $Q^*(8) = 5.35$ on 5 degrees of freedom, $p = 0.375$; $Q^*(16)$: $p = 0.525$
  - every p-value in the chart is above 0.12; with 8 instead of 5 degrees of freedom, p would be 0.720, too reassuring
- Not Normal: Jarque–Bera 39.0, $p < 0.001$; skewness -0.75, kurtosis 5.61
  - the largest residual: -11.4 in 2020Q2, -3.89 standard deviations: the pandemic lockdown
- Squared residuals: Ljung–Box $p = 0.779$, no volatility clustering at this frequency
- Verdict: the MA(3) captures the dependence; the Normal intervals understate the risk of recessions

Step 4: forecasts



- MA(3) (BIC) and AR(1) forecasts for 2026Q3–2028Q2, with 95% intervals

TSA_ch2_gdp_case

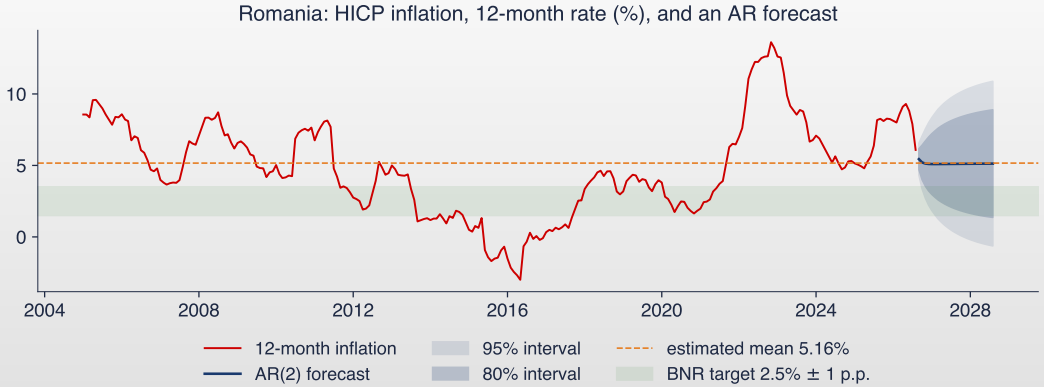
Interpreting the GDP forecasts

- MA(3): -0.14, 1.81, 2.77, then the mean 3.17% from the fourth quarter on
  - exactly the MA(q) rule: after $q = 3$ steps the model only knows the mean
- AR(1): a smooth return to its mean, 2.99% after 8 quarters (mean 3.12%)
  - the two models agree in the short run and differ in how fast they forget the recent slowdown
- Intervals: $[-6.1, 5.9]$ one quarter ahead, $[-6.3, 12.7]$ from four quarters on: ± 6.0 and ± 9.5 percentage points
- A univariate model cannot anticipate a turning point; it tells us how unusual the next value would be

Recap: Romanian GDP growth

- Annual growth of quarterly GDP: ACF cuts off after lag 3, MA(3) by BIC, ARMA(1,3) by AIC
- The MA(3) is the overlap of four quarters: the data structure explains the model
- White-noise residuals but fat tails (2020): intervals are too narrow in crises
- Forecasts reach the mean after three quarters

Romanian inflation: an AR(2) forecast



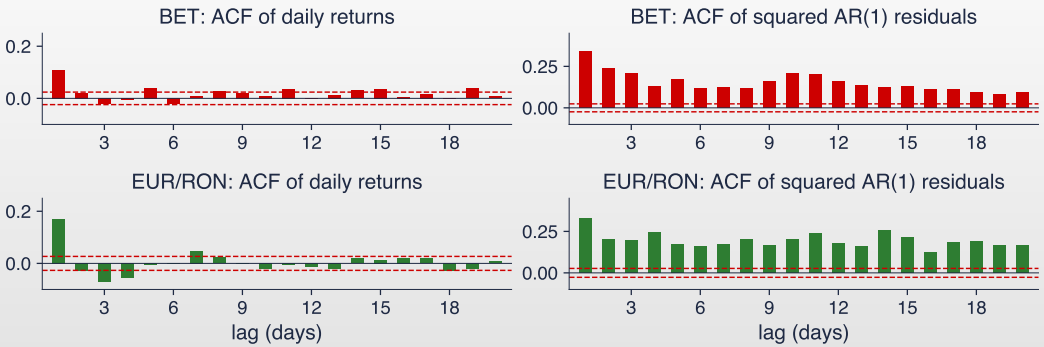
- HICP (harmonised index of consumer prices, Eurostat), 12-month rate in %, 260 months to August 2026; AR order chosen by BIC among $p \leq 6$

TSA_ch2_real_series

Interpreting the inflation model

- BIC: 495.7 for AR(1), 469.7 for AR(2), 473.4 for AR(3): AR(2)
  - $\hat{\phi}_1 = 1.320$ (0.057), $\hat{\phi}_2 = -0.344$ (0.055); $\hat{\phi}_1 + \hat{\phi}_2 = 0.976$, largest inverse root 0.963
- Very close to a unit root: is inflation stationary at all? The test is in Chapter 3
  - the estimated mean, 5.16% (se 1.35), is the average of 2005–2026, above the BNR target of 2.5% $\pm$ 1 p.p.
- Forecast: from 6.09% to 5.44% next month and 5.13% in August 2028; 95% interval $[-0.6, 10.9]$
  - the model reverts to its historical mean, not to the target: it knows nothing about monetary policy
- Diagnostics fail: Ljung–Box $Q^*(12) = 71.6$, p < 0.001: correlation at lag 12 (base effects of the 12-month rate), a job for Chapter 4

Daily returns: BET and EUR/RON



- Left: ACF of daily log returns; right: ACF of the squared residuals of an AR(1)

TSA_ch2_real_series

Interpreting the AR(1) for daily returns

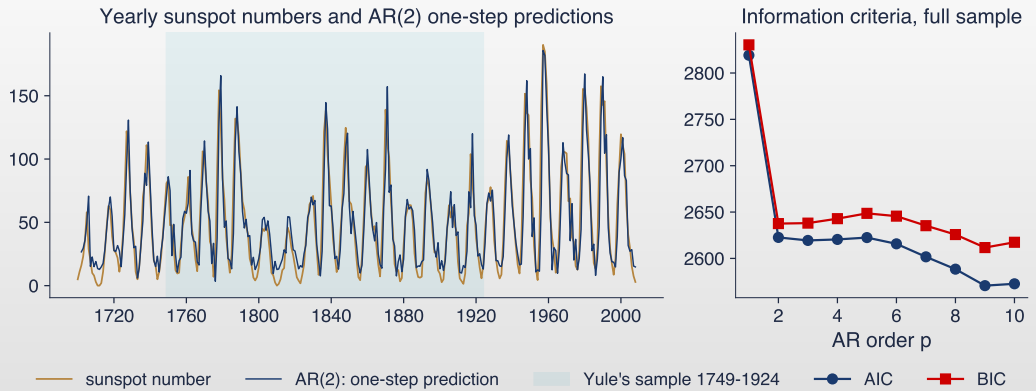
	BET	EUR/RON
Observations	6 670	5 352
$\hat{\phi}$ (se OPG)	0.109 (0.005)	0.170 (0.006)
t-ratio	20.9	30.6
R^2 (%)	1.2	2.9
LB $Q^*(10)$ residuals, 9 df (p)	29.7 (<0.001)	61.5 (<0.001)
LB $Q^*(10)$ squared residuals	2496	2351
Kurtosis of residuals	14.0	17.0
BIC choice, $p, q \leq 2$	ARMA(1,0)	ARMA(2,1)

- se: OPG; with heavy tails (kurtosis 14.0 and 17.0) they are too small, and curvature or robust se are several times larger, yet $\hat{\phi}$ stays significant
- Highly significant $\hat{\phi}$, but R^2 of 1–3%: **statistically** real, **economically** small after trading costs
- Positive $\hat{\phi}$: slow price adjustment (BET), a managed exchange rate (EUR/RON)
- Squared residuals and kurtosis: the mean is modelled, the variance is not; ARMA + GARCH in Chapter 5

Case study: Yule (1927) and the sunspots

- Yule (1927) fitted the first autoregression: Wolfer's yearly sunspot numbers, 1749–1924
  - until then, cycles were modelled as sums of sine waves (periodogram)
  - Yule: a "pendulum bombarded by peas": a damped oscillator kicked by random shocks
- Our least squares AR(2) on the same years (176 observations, statsmodels data):
  - $\hat{\phi}_1 = 1.336$, $\hat{\phi}_2 = -0.650$: complex roots, modulus of the inverse roots 0.806
  - implied period $2\pi / \arccos(\hat{\phi}_1 / (2\sqrt{-\hat{\phi}_2})) = 10.57$ years: the solar cycle, from two coefficients
- Walker (1931) generalised the equations to AR(p): the Yule–Walker equations
  - the start of parametric time series analysis

The sunspots: AR(2) and the information criteria



⊡ Yearly sunspot numbers, 1700–2008 (309 years); AR(p) by MLE on the full sample

TSA_ch2_real_series

Interpreting the sunspot models

- Full-sample AR(2): $\hat{\phi} = (1.391, -0.689)$, close to Yule's; it explains 83% of the variance one year ahead
  - the one-step predictions follow the cycle, but lag behind at the peaks
- AIC and BIC both choose $p = 9$ over $p \leq 10$
  - the cycle is asymmetric (fast rise, slow decay), which a linear AR(2) cannot reproduce
- Lesson: two parameters capture the essence; more lags capture details; nonlinearity is beyond ARMA

Case study: forecasting competitions

- Makridakis and Hibon (2000): the M3 competition, 3003 series (yearly, quarterly, monthly, other), 24 methods
  - forecasts compared out of sample, on several accuracy measures and horizons
- Conclusions confirmed from the earlier M-competitions
  - statistically sophisticated methods do not necessarily forecast better than simple ones
  - the ranking depends on the accuracy measure and on the horizon; combinations of methods do well
- Consequences for ARMA
  - the Box–Jenkins loop must end with an out-of-sample comparison against simple benchmarks (mean, last value, exponential smoothing from Chapter 0)
  - Seminar 2, C1 runs this comparison for Romanian inflation

Recap: More real series

- Inflation: AR(2) near a unit root; reverts to the historical mean; lag-12 residual correlation
- Daily returns: significant but tiny AR effects; the variance needs GARCH
- Sunspots: Yule's AR(2) gives the 11-year cycle from complex roots
- Forecasting competitions: always compare with simple benchmarks

Possible contribution of AI

- **Code**: a first draft of a loop that fits ARMA(p, q) on a grid, tabulates AIC and BIC, and runs the residual tests
- **Explanation**: a second derivation of the ACF of an ARMA(1,1), or of why the PACF of an MA decays
- **Exploration**: the same Box–Jenkins loop on many series (GDP growth of all EU countries) to compare persistence
- Example prompt
  - `Write Python code with statsmodels that fits ARMA(p,q), p,q <= 3, to Romanian annual GDP growth from Eurostat, reports AIC, BIC and the Ljung-Box p-value with model_df = p + q, and forecasts 8 quarters from the BIC model.`

Necessary checks

- The sign convention of θ and ϕ in the software ($\theta(L) = 1 + \theta L$ in statsmodels, $1 - \theta L$ in some books)
- Intercept or mean: the `const` of `ARIMA` is the mean μ, not c
- Ljung–Box degrees of freedom on residuals ($m - p - q$), and that all models are compared on the same sample
- Roots of both polynomials, not the size of single coefficients
- That the series is stationary before an ARMA is fitted (Chapter 3), and that forecasts are compared with simple benchmarks
- Every cited reference: it must exist; check the DOI

Key takeaways

- ARMA(p, q) replaces the infinite Wold sum by a ratio of two short polynomials, $\theta(L)/\phi(L)$
- Stationarity: roots of $\phi(z)$ outside the unit circle; invertibility: roots of $\theta(z)$ outside; no common factors
- ACF cuts off: MA; PACF cuts off: AR; both decay: ARMA
- Estimate by MLE, choose by AIC/BIC, check residuals with Ljung–Box on $m - p - q$ degrees of freedom
- Forecasts revert to the mean; intervals widen to $\pm 1.96\sqrt{\gamma(0)}$ and ignore parameter uncertainty
- Real data: GDP growth is an MA(3) by construction; inflation is near a unit root; returns need GARCH

Key formulas

Quantity	Formula
ARMA(p,q)	$\phi(L)(X_t-\mu)=\theta(L)\varepsilon_t, \quad \psi(L)=\theta(L)/\phi(L)$
AR(1)	$\mu=c/(1-\phi), \quad \gamma(0)=\sigma^2/(1-\phi^2), \quad \rho(h)=\phi^h$
AR(2)	$\rho(1)=\phi_1/(1-\phi_2), \quad \rho(h)=\phi_1\rho(h-1)+\phi_2\rho(h-2)$
MA(1)	$\rho(1)=\theta/(1+\theta^2), \quad \rho(h)=0,\ h\geq 2$; invertible $\lvert\theta\rvert<1$
ARMA(1,1)	$\psi_j=\phi^{j-1}(\phi+\theta), \quad \rho(h)=\phi\rho(h-1),\ h\geq 2$
Yule–Walker	$\hat{\phi}=\hat{R}^{-1}\hat{\rho}, \quad \hat{\sigma}^2=\hat{\gamma}(0)(1-\hat{\phi}^\top\hat{\rho})$
Criteria	$\mathrm{AIC}=-2\ln L+2k, \quad \mathrm{BIC}=-2\ln L+k\ln T$
Residuals	$Q^*(m)\sim\chi^2(m-p-q)$
Forecast	$\hat{X}_{T+h\mid T}\pm 1.96\,\sigma\sqrt{\sum_{j=0}^{h-1}\psi_j^2}$

Self-assessment

- **Question**: is $X_t = 1.2X_{t-1} - 0.32X_{t-2} + \varepsilon_t$ stationary?
  - **Answer**: $1 - 1.2z + 0.32z^2 = (1 - 0.8z)(1 - 0.4z)$, roots 1.25 and 2.5, both outside the circle: yes
- **Question**: the ACF of a series has one spike $\hat{\rho}(1) = -0.45$ and the PACF decays. Which model?
  - **Answer**: MA(1) with $\theta < 0$; solve $\theta/(1 + \theta^2) = -0.45$ and keep the invertible root
- **Question**: an ARMA(2,1) leaves $Q^*(10) = 15.2$ in its residuals. Reject at 5%?
  - **Answer**: degrees of freedom $10 - 3 = 7$, $\chi^2_{0.95}(7) = 14.07 < 15.2$: reject; with 10 degrees of freedom (critical 18.31) one would wrongly accept
- Next: Chapter 3, unit roots and ARIMA models: what to do when a root sits on the unit circle

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